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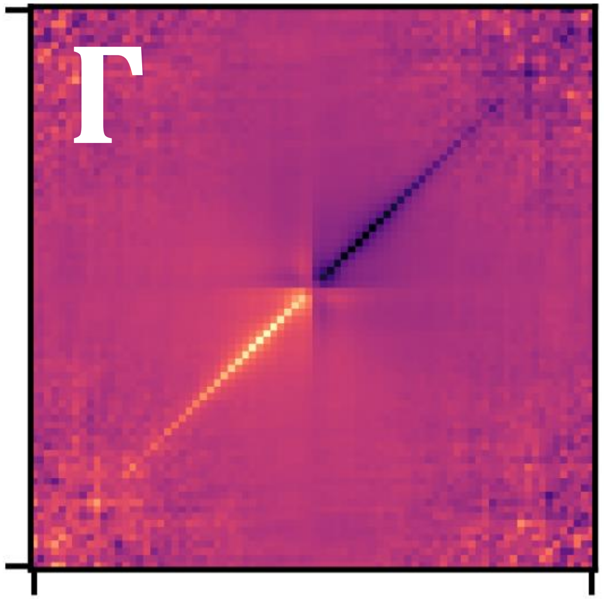

Multi-orbital dynamical vertex approximation (DΓA)

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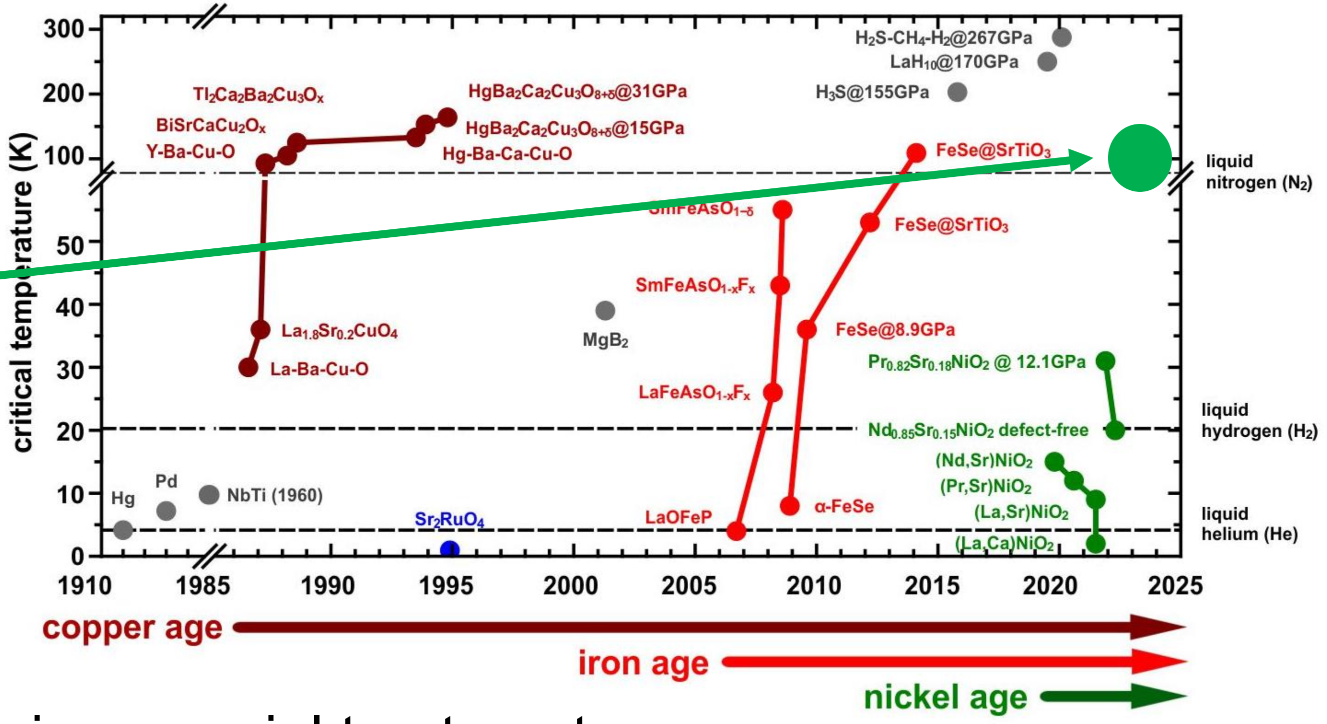
$La_3Ni_2O_7$
@ 14 GPa

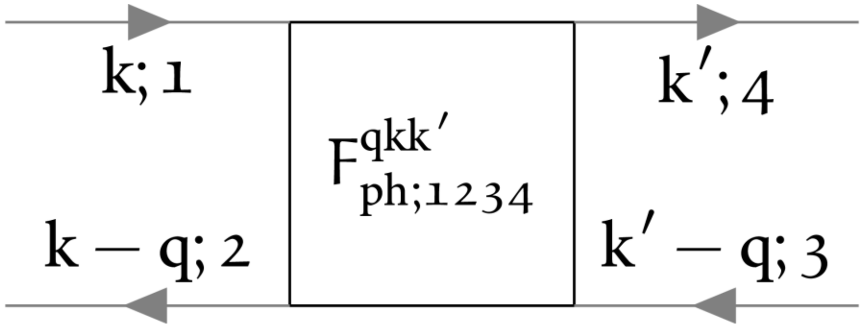


Motivation

- Unconventional superconductivity (e.g. in cuprates or nickelates) requires special treatment DMFT → DΓA [1]
- DΓA & superconductivity only implemented for one orbital [2, 3]
- Many correlated materials (e.g. $La_3Ni_2O_7$) need a multi-orbital description [4]
- Calculations down to low temperatures → vertex asymptotics [5]

[1] G. Rohringer et. al., Rev. Mod. Phys. 90 (2018)
[2] P. Worm, Zenodo (2023)
[3] A. Galler et al., Comput. Phys. Commun. 245 (2019)
[4] H.Sun et al., Nature 621 (2023)
[5] M. Kitatani et al., J. Phys. Mater 5 (2022)

Implementation details [6]



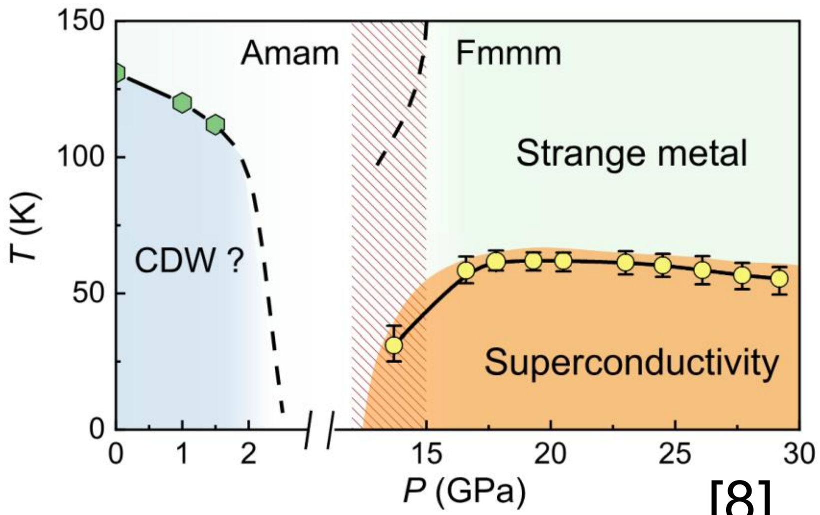
- From single- to multi-orbital:
  - Green's functions & self-energies gain two orbital indices: $G^{k} \rightarrow G^{k}_{12}$
  - Two-particle vertex functions gain four orbital indices: $F^{qkk'} \rightarrow F^{qkk'}_{1234}$
  - Memory requirements ↑ and computation time ↑
- Multiplications include orbital contractions, e.g. $C^{qkk'}_{1234} = A^{qkk_1}_{12ab} B^{qk_1k'}_{ba34}$
- Parallelized code that takes advantage of system symmetries (e.g. full → irreducible Brillouin zone, frequency symmetries, …)

[6] https://github.com/Julpe/moLDGA

Benchmarks

- Agreement with single-band code up to numerical precision [1] ✓
- Correct decoupling for disconnected two-band tests ✓
- Performance: ~12 h → ~30 min for low-temperature data compared to [1]

Outlook



[8]

- Testing limits of the code
- Benchmarks against multi-orbital calculations performed by [2, 7] ($SrVO_3$)
- Calculation of realistic materials (especially $La_3Ni_2O_7$)

[1] P. Worm, Zenodo (2023)
[2] A. Galler et al., Comput. Phys. Commun. 245 (2019)
[7] J. Kaufmann et. al., Phys. Rev. B 103 (2021)
[8] Y. Zhang et al., Nature 20 (2024)

Summary

- Motivation
  - Study realistic multi-orbital materials (e.g. $La_3Ni_2O_7$) with DΓA
- Implementation
  - Additional orbital degrees of freedom
  - Taking advantage of system symmetries
- Successful execution of benchmarks
  - Single- and two-orbital calculations

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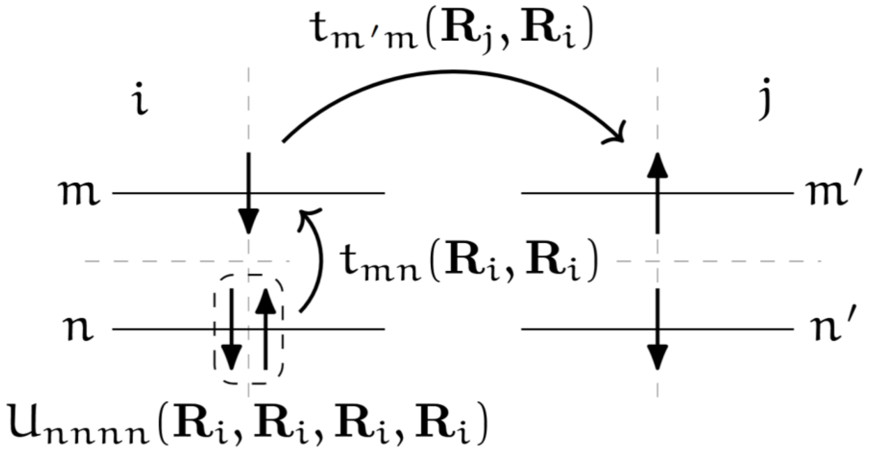

Model

- Multi-orbital Hubbard Model

$$
\begin{aligned}
\mathcal{H} = &- \sum_{\substack{\mathbf{R}\mathbf{R}' \\ mm',\sigma}} t_{mm'}(\mathbf{R}, \mathbf{R}') \hat{c}^{\dagger}_{\mathbf{R}m\sigma} \hat{c}_{\mathbf{R}'m'\sigma} \\
&+ \frac{1}{2} \sum_{\substack{\mathbf{R}_1\mathbf{R}_2\mathbf{R}_3\mathbf{R}_4 \\ ll'mm' \\ \sigma\sigma'}} U_{lmm'l'}(\mathbf{R}_1, \mathbf{R}_2, \mathbf{R}_3, \mathbf{R}_4) \hat{c}^{\dagger}_{\mathbf{R}_1 l\sigma} \hat{c}^{\dagger}_{\mathbf{R}_3 m'\sigma'} \hat{c}_{\mathbf{R}_4 l'\sigma'} \hat{c}_{\mathbf{R}_2 m\sigma}
\end{aligned}
$$